

Federated Unlearning Framework for Privacy-Preserving Inter-Hospital Healthcare Networks in Smart Cities

Amani Albarazi

Effat University

College of Engineering

Computer Science Department

Jeddah, Saudi Arabia

Aalbarazi@effat.edu.sa

Farha Hussaini

Effat University

College of Engineering

Computer Science Department

Jeddah, Saudi Arabia

fayhussaini.edu.sa

Abstract

Federated learning (FL) allows hospitals to collaboratively train models without sharing patient data, but existing unlearning techniques such as FATS and IFU face challenges including sensitivity to non-IID data, high computational cost, and limited support for multi-client deletion requests. This work proposes a hybrid federated unlearning framework combining client-level FATS and patient-level IFU to efficiently remove the influence of specific hospitals or patients. The system integrates granularity detection, adaptive optimization, and verification modules to ensure privacy, stability, and regulatory compliance. A use-case on a benchmark healthcare dataset demonstrates patient-level unlearning without full retraining while maintaining model performance. The framework addresses key gaps in current methods, offering precise, scalable, and auditable unlearning for heterogeneous hospital networks.

Keywords: Federated Learning, Unlearning, Influence Functions, FATS, Healthcare, Privacy.

1 Introduction

The integration of artificial intelligence (AI) into healthcare has been significantly accelerated by *federated learning (FL)*, a paradigm that enables multiple hospitals and healthcare institutions to collaboratively train machine learning models without exchanging raw patient data. By keeping sensitive patient information local, FL not only protects privacy but also allows institutions to leverage collective knowledge, improving predictive analytics, diagnostic accuracy, and resource optimization across the healthcare network (9). Despite these advantages, a major challenge remains: guaranteeing the *right to be forgotten*, which requires the reliable removal of a patient's contribution from a trained model without compromising model performance.

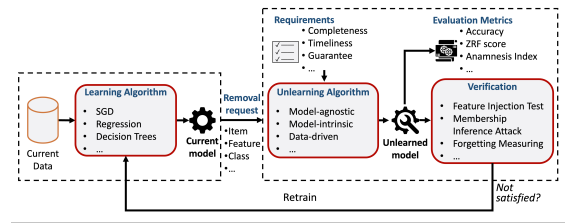


Figure 1: A Typical Machine Unlearning Process

Traditional retraining approaches are computationally expensive, time-consuming, and often impractical in large-scale inter-hospital networks. To address this limitation, *federated unlearning (FU)* has emerged as a critical solution, providing methods to efficiently erase specific data contributions at either the client or sample level while maintaining the integrity and predictive power of the global model (9). Advanced unlearning techniques, such as TV-stable federated algorithms (FATS) for exact unlearning, influence-function-based methods, and Model-Contrastive Federated Unlearning (MCFU), offer theoretical guarantees and practical mechanisms to remove data influence in a communication-efficient manner (4).

Beyond computational concerns, ensuring that unlearning actions are verifiable and cannot be falsely claimed is essential for maintaining trust among patients and healthcare institutions, particularly in sensitive domains such as medical imaging and electronic health records (4). In smart-city healthcare networks, where hospitals increasingly rely on shared AI models for diagnosis, predictive healthcare, and operational management, a robust federated unlearning framework becomes indispensable. Such a framework enables hospitals to honor patients' requests under the "right to be forgotten," preserves model utility and generalization, minimizes computation and communication overhead for practical deployment, and incorporates security measures to guarantee that unlearning is genuine and transparent (?). By addressing these

interrelated challenges, federated unlearning not only ensures compliance with data protection regulations but also fosters trust, accountability, and broader adoption of AI-driven healthcare solutions.

1.1 Importance of the Topic

In smart-city healthcare networks, hospitals increasingly rely on collaboratively trained AI models for diagnosis, predictive healthcare, and resource management. However, the inability to reliably remove patient data can violate privacy regulations, such as the GDPR, and erode patient trust. This work addresses these challenges by proposing a federated unlearning framework that enables hospitals to process “right to be forgotten” (RTBF) requests while maintaining model accuracy, generalization, and privacy. The system also supports verifiable and secure unlearning, promoting trustworthy collaborative AI. In terms of global goals, this framework contributes to **SDG 3 (Good Health and Well-Being)** by improving healthcare delivery, **SDG 9 (Industry, Innovation, and Infrastructure)** by providing scalable and efficient AI infrastructure, and **SDG 16 (Peace, Justice, and Strong Institutions)** by ensuring ethical, compliant, and accountable AI practices.



Figure 2: Sustainable Development Goals

1.2 Problem Definition and Scope

The integration of a Federated Unlearning framework into federated learning systems within inter-hospital networks presents several challenges. The framework must enable hospitals to efficiently remove specific patient data from shared models without compromising model accuracy or generalization. It should ensure compliance with data protection regulations, such as the GDPR, supporting patients’ right to be forgotten, while maintaining computational and communication efficiency suitable for large-scale deployment. Additionally, the framework must incorporate security mechanisms to guarantee that unlearning actions are verifiable and cannot be misrepresented, thereby fostering trust in collaborative AI systems across healthcare networks.

Research Question: How can a Federated Unlearning framework be effectively integrated into federated learning systems within inter-hospital networks to ensure privacy, regulatory compliance, and model accuracy while supporting patients’ right to be forgotten?

2 federated unlearning literature review

Federated learning (FL) has gained prominence in hospital environments because it enables collaborative model development without exposing raw patient data. As healthcare systems increasingly adopt FL for radiology, pathology, and electronic medical records, modern privacy regulations such as GDPR and national health data laws require support for verifiable deletion of patient or institutional data contributions. This necessity has motivated recent research on federated unlearning (FUL), which aims to remove the influence of specific clients or samples from trained models without full retraining.

Early FUL studies extended the standard FedAvg algorithm by attempting to reverse historical client contributions during model aggregation. These rollback-based techniques were tested in multi-hospital medical imaging collaborations, allowing institutions to join and leave dynamically (7; 8). Although conceptually simple, these approaches require storing large logs of historical gradients and do not provide strong guarantees that deleted clients’ contributions are fully removed.

More recent work introduced training-stabilized unlearning. Federated Averaging with Training Stabilization (FATS) represents a major advancement by incorporating Lipschitz-bounded updates and variance-control constraints directly into the FL workflow (9). These constraints make the global model inherently more reversible, enabling deletion through a small deterministic correction rather than full retraining. Studies show that FATS supports both client-level and patient-level unlearning with low communication overhead and provable guarantees—properties critical for cross-hospital networks (9; 10). Empirical evaluations indicate that FATS outperforms rollback-based methods in healthcare FL, especially in large-scale radiology tasks.

Parallel to stabilization-based unlearning, influence-function-based unlearning (IFU) has been proposed to support fine-grained patient-level deletion. IFU estimates how each training sample

affects the model parameters and then subtracts this influence to approximate unlearning (11; 12). This approach is computationally efficient and particularly beneficial for healthcare datasets involving mixed imaging and structured patient records. However, IFU relies on convexity and smoothness assumptions that may not fully hold for deep neural networks, leading to approximate rather than exact removal.

For scenarios where entire hospitals, departments, or medical devices must be removed from the federation, membership- or client-focused unlearning (MCFU) has been explored. These techniques use aggregated-update rollback or model-difference reconstruction to eliminate the departing client's contributions (13; 14). MCFU is easier to integrate into existing federated healthcare pipelines but lacks strong theoretical erasure guarantees and is not designed for patient-level deletion.

Recently, the importance of certified unlearning has grown in healthcare due to audit and compliance requirements. Certified federated unlearning uses tools such as Bregman-divergence-based verification, differential privacy, and cryptographic attestations to mathematically prove that the requested data no longer influences the model (15; 16). Although computationally expensive, these methods provide regulatory-grade assurance.

Across all approaches, FATS and IFU stand out as the most effective techniques for hospital federated systems. FATS offers the best balance of efficiency, accuracy, and provable guarantees for both client-level and sample-level unlearning, while IFU provides ultra-efficient fine-grained patient deletion. Together, these methods represent the state-of-the-art FUL solutions for healthcare applications from 2022 to 2025.

3 gaps and findings

Despite progress in federated unlearning, several challenges remain before such methods can be reliably deployed in hospital environments. First, existing client-level unlearning techniques frequently exhibit instability when faced with the highly non-IID data distributions typical of multi-hospital networks. Rollback-based approaches in particular suffer from divergence and loss of model quality, indicating the need for more stable mechanisms; in this regard, FATS currently represents the most robust and effective solution reported in the literature. Second, fine-grained patient-level unlearning

remains computationally demanding or only approximately correct for modern deep clinical models. Many proposed methods cannot guarantee precise removal of individual patient contributions, whereas influence-function-based unlearning (IFU) remains the most accurate and efficient technique available for sample-level deletion. Third, current federated unlearning frameworks lack comprehensive support for the operational realities of hospital systems, including multimodal data (imaging, EHR, sensor streams), dynamic site participation, and regulatory auditing requirements. These limitations collectively highlight that FATS and IFU are presently the strongest and most practically suitable unlearning techniques for hospital-centric federated learning.

4 Core Technique

Federated Unlearning (FU) has emerged as a critical extension of Federated Learning (FL) in response to increasing privacy demands and the “Right to Be Forgotten” (RTBF) established in major regulations such as the GDPR, CCPA, and APPI (24). In conventional machine learning systems, removing a user's data simply deletes the stored samples, yet the influence of that data remains embedded in the trained model's parameters. To address this challenge, the field of Machine Unlearning (MU) was introduced, enabling models to eliminate the effect of specific data without full retraining (22).

FU extends MU to distributed environments where multiple clients—such as hospitals, research institutions, or edge devices—collaboratively train a global model without sharing raw data. Despite this decentralization, each client's updates still encode information derived from its local data. FU enables the system to remove this influence by supporting both client-level unlearning (erasing the entire contribution of a client) and data-level unlearning (forgetting selected samples within a client) (31). The goal is to ensure that the updated global model behaves as if the forgotten data never participated in training, without violating privacy constraints or incurring costly full retraining.

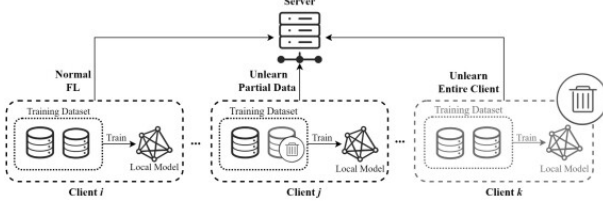


Figure 3: Core concept of Federated Unlearning (FU), showing normal FL operation, partial data unlearning, and full client unlearning within a multi-client federated setup.

4.1 FATS: Shard-Based Federated Unlearning

FATS (Federated Averaging with Total Variation Stability) allows exact and approximate unlearning. Client datasets are partitioned into shards, and each shard trains a local model. Total Variation (TV) stability ensures that removing a single data point changes the model output by at most :

$$D_{TV}(M(D), M(D \setminus \{x\})) \leq \epsilon.$$

When a deletion request occurs, only the affected shard is retrained while others remain intact:

$$w' = \frac{1}{S} \left(w'_k + \sum_{j \neq k} w_j \right),$$

where w'_k is the retrained shard and S is the total number of shards. Gradient clipping ensures bounded sensitivity:

$$g_s^t = \text{clip}(\nabla L_s(w^t), C), \quad (1)$$

$$w_s^{t+1} = w^t - \eta g_s^t, \quad (2)$$

$$w^{t+1} = \frac{1}{S} \sum_{s=1}^S w_s^{t+1}. \quad (3)$$

Algorithm 1 FATS Shard-Based Federated Unlearning

Require: Shards $\{D_1, \dots, D_S\}$, learning rate η , clip constant C , rounds T

- 1: Initialize global model w_0
- 2: **for** $t = 0$ to T **do**
- 3: **for** $s = 1$ to S **do**
- 4: $g_s^t \leftarrow \text{clip}(\nabla L_s(w^t), C)$
- 5: $w_s^{t+1} \leftarrow w^t - \eta g_s^t$
- 6: **end for**
- 7: $w^{t+1} = \frac{1}{S} \sum_{s=1}^S w_s^{t+1}$
- 8: **end for**
- 9: Identify affected shard k
- 10: $D'_k = D_k \setminus D_{\text{del}}$
- 11: $w'_k = \text{Train-TV-Stable}(D'_k)$
- 12: $w' = \frac{1}{S} (w'_k + \sum_{j \neq k} w_j)$
- return** w'

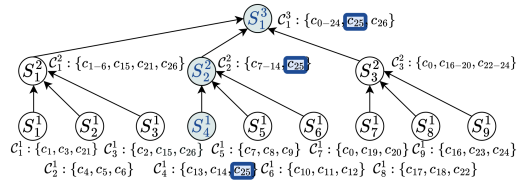


Figure 4: Example FATS Shard-Based Federated Unlearning Node Representation

4.2 Influence Function-Based Unlearning

Influence Function-Based Unlearning (IFU) provides a theoretically grounded method to estimate the effect of individual training samples on a model's parameters, enabling selective unlearning without retraining the entire model. This is particularly relevant for applications in *privacy compliance*, *federated learning*, and *dynamic model updates*.

For a model trained via empirical risk minimization on a dataset $\mathcal{D} = \{z_1, z_2, \dots, z_n\}$, the optimal parameters are:

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n L(z_i, \theta),$$

where $L(z_i, \theta)$ is the loss associated with sample z_i and model parameters θ .

The influence of a single data point z_j on the model parameters is approximated by:

$$I(z_j) = -H_{\hat{\theta}}^{-1} \nabla_{\theta} L(z_j, \hat{\theta}),$$

where $H_{\hat{\theta}}$ is the Hessian of the total loss with respect to θ evaluated at $\hat{\theta}$, and $\nabla_{\theta} L(z_j, \hat{\theta})$ is the gradient of the loss with respect to θ at z_j . Using this influence, the updated model parameters after removing z_j are:

$$\hat{\theta}_{-j} \approx \hat{\theta} + \frac{1}{n} I(z_j).$$

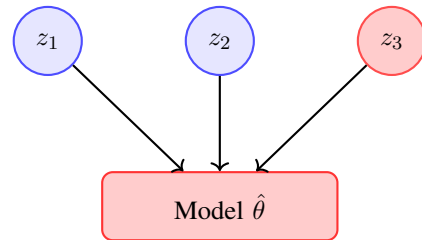


Figure 5: Illustration of Influence Function-Based Unlearning.

4.2.1 FedU via User-Side Influence

A practical application of IFU is the **FedU framework** for federated learning. Each client estimates the influence of data points to be removed and subtracts it from the global model, a process termed *Influence Approximation Forgetting (IAF)*. This is illustrated by the black arrows in Figure 8. After influence subtraction, local retraining is performed on the remaining dataset (blue arrows) to preserve model utility. Adaptive optimization is applied to minimize both the influence subtraction loss and the local retraining loss concurrently.

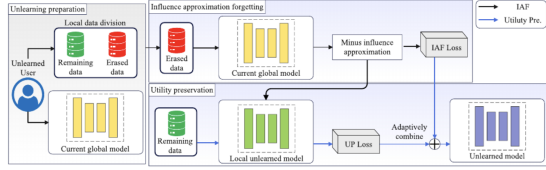


Figure 6: FedU via user-side influence approximation forgetting. Black arrows represent influence subtraction, and blue arrows represent local retraining for utility preservation.

4.2.2 IFU Example for Simple Dataset

To demonstrate IFU, consider a linear regression problem with three data points:

x	y
1	2
2	4
3	6

The model $y = \theta x$ is trained using the mean squared error loss:

$$L(z_i, \theta) = \frac{1}{2}(y_i - \theta x_i)^2$$

The optimal parameter for the full dataset is $\hat{\theta} = 2$. Suppose we wish to remove the point $z_3 = (3, 6)$. Using the influence function:

- Gradient: $\nabla_{\theta} L(z_3, \hat{\theta}) = -x_3(y_3 - \hat{\theta}x_3) = 0$
- Hessian: $H_{\hat{\theta}} = \sum_i x_i^2 = 14$

Thus, $I(z_3) = 0$, and the updated parameter after removal is:

$$\hat{\theta}_{-3} \approx \hat{\theta} + \frac{1}{3}I(z_3) = 2$$

This simple example demonstrates that IFU can efficiently “forget” a data point without retraining, while preserving model parameters when the point has minimal influence.

Algorithm 2 Influence Function-Based Unlearning for Logistic Regression (PyTorch)

Require: Dataset (X, y) , trained logistic regression model f_{θ} , patient index i to remove, learning rate η , LiSSA iterations T

Ensure: Updated model parameters $\hat{\theta}_{-i}$ with patient i influence removed

1: Standardize features: $X \leftarrow \text{StandardScaler}(X)$

2: Convert X, y to torch tensors

3: Train logistic regression model f_{θ} with BCE loss using SGD

4: Select patient to remove: $(x_i, y_i) \leftarrow X[i], y[i]$

5: Compute per-patient gradient: $g_i = \nabla_{\theta} L(f_{\theta}(x_i), y_i)$

6: Define Hessian-vector product function $Hv = \nabla^2 L(f_{\theta}(x_i), y_i)$ using autograd

7: Approximate $H^{-1}g_i$ via fixed-point iteration:

$$v \leftarrow g_i$$

8: **for** $t = 1$ to T **do**

9: $Hv \leftarrow \text{hessian_vector_product}(f_{\theta}, \text{loss over dataset}, v)$

10: $v \leftarrow v - \eta(Hv - g_i)$

11: **end for**

12: $h_inv_g \leftarrow v$

13: Compute parameter update: $\Delta\theta \leftarrow \frac{1}{n}h_inv_g$

14: Update model parameters: $\theta \leftarrow \theta + \Delta\theta$

15: **return** Updated model $f_{\hat{\theta}_{-i}}$

4.3 Influence Function-Based Unlearning Experiment

To demonstrate the practical application of *Influence Function-Based Unlearning (IFU)*, we conducted an experiment using the *Breast Cancer Wisconsin dataset*, a widely-used medical dataset containing 30 numerical features describing cell nuclei and a binary target indicating benign (1) or malignant (0) tumors. The dataset was standardized and converted into PyTorch tensors for efficient computation.

A logistic regression model was trained on the full dataset using binary cross-entropy loss and stochastic gradient descent for 1000 epochs. After training, we selected patient 10 (the 11th sample) for unlearning. The per-patient gradient with respect to model parameters was computed, and the Hessian-vector product was approximated using a fixed-point iteration (LiSSA) to estimate $H^{-1}g_i$. The model parameters were then updated according to the influence function formula, effectively removing the selected patient’s influence while preserving the rest of the model’s knowledge. The gradient vector had shape [31], corresponding to 30 weights plus one bias term.

The full implementation of this procedure in PyTorch, including data preprocessing, model training, gradient computation, Hessian-vector approximation, and parameter update, is provided in **Appendix ??**. results shown in fig 5

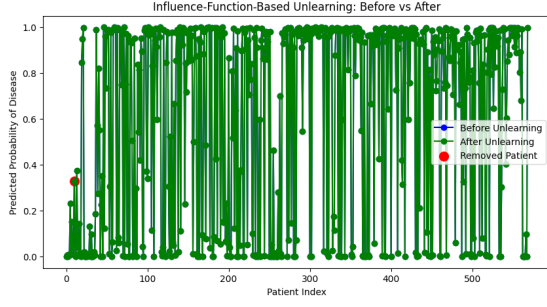


Figure 7: Influence-Function-Based Unlearning: Before vs After

4.3.1 Summary

Influence Function-Based Unlearning provides a computationally efficient, theoretically grounded approach for selective data removal. When combined with local retraining or adaptive optimization, IFU achieves a balance between *effective unlearning* and *utility preservation*, making it suitable for federated learning, privacy-sensitive applications, and dynamic model management.

5 Survey of Recent Advances

Recent work on Federated Unlearning has explored several techniques aimed at removing client data without retraining the entire model. Early approaches relied on knowledge-distillation-based unlearning, where a new model is trained only using the clients that remain in the system. This reduces retraining cost but still leaves part of the removed client’s influence, especially when hospital data is highly non-IID (35). Another line of research introduced server-side gradient filtering, which removes or ignores old updates from the client being forgotten; however, the global model still keeps some of that client’s effect because earlier updates were already mixed into aggregation (36). A third direction used partial layer retraining, where only selected layers influenced by the client are updated, but this becomes unstable on real medical datasets with strong variation between hospitals (37). Overall, these earlier methods achieved only partial forgetting and were not reliable or scalable enough for real inter-hospital networks.

5.1 Federated Averaging with Total Variation Stability (FATS)

Among the most significant advancements in this field is Federated Averaging with Total Variation Stability (FATS). FATS extends the classical Federated Averaging (FedAvg) algorithm by embedding

a property known as Total Variation (TV) Stability, which measures how sensitive the global model is to small perturbations or removals in client data. This stability constraint ensures that the training process does not overfit to any specific hospital, enabling exact unlearning—the ability to remove a client’s or patient’s contribution and recover a model identical to one retrained from scratch. This directly addresses limitations in earlier FU methods such as FedEraser, gradient-compensation approaches, and influence-function unlearning, all of which provide only approximate forgetting and lack formal guarantees of correct deletion (26).

The original FATS framework was validated through experiments on six benchmark datasets, where it consistently outperformed prior FU methods in accuracy, stability, and communication efficiency (26). Beyond these results, subsequent research conducted deeper analyses of how FATS behaves in realistic federated environments. For instance, evaluations under heterogeneous data distributions—common in inter-hospital networks—revealed that while FATS maintains strong theoretical guarantees, its performance may be challenged when hospitals have highly non-IID data due to differences in imaging protocols, patient demographics, or sample sizes (23). Survey studies further classify FATS as the foundational stability-based FU method and highlight its influence on how reversibility, forgetting accuracy, and compliance should be measured in distributed medical systems (30). Other researchers have reused FATS’s TV-stability principle to design architecture-specific unlearning mechanisms, extend stability constraints to convolutional and attention-based models, and redefine evaluation metrics for certifiable unlearning (28). Practical FU methods—such as consistency-based and server-side model-surgery approaches—also benchmark their deletion correctness against FATS, treating it as the gold standard for measuring whether unlearning matches the retraining-from-scratch ideal (22).

In the context of inter-hospital healthcare networks, FATS’s guarantees are particularly valuable. It supports both client-level unlearning, allowing a hospital to withdraw its data entirely, and sample-level unlearning, enabling compliance with patient-level Right to Be Forgotten (RTBF) requests. Medical federated learning studies frequently reference FATS as the conceptual definition of “complete

forgetting,” even when implementing alternative unlearning techniques (3). This reflects a broader trend: FATS is not only a standalone algorithm but also a theoretical framework that shapes how privacy, deletability, auditability, and compliance are defined in federated healthcare systems. By providing verifiable and auditable forgetting, FATS bridges the gap between theoretical unlearning guarantees and the operational requirements of real-world hospitals, making it a foundational model for secure, regulation-compliant, and trustworthy inter-hospital collaborations.

5.2 Influence function-based unlearning

Machine unlearning, defined as the selective removal of specific training data’s influence from a trained model, has become a critical area of research in the context of privacy, fairness, and data integrity (18). Full retraining of models after data deletion is computationally prohibitive, particularly for large-scale and deep neural network architectures, motivating the development of approximate unlearning techniques (20). Influence-function-based unlearning provides a principled framework to estimate the impact of individual data points on the learned model parameters and to approximate the effect of deletion without full retraining (2).

Influence-function-based unlearning has also been extended to the removal of entire features or labels, rather than only individual instances (18). Consider a model parameterized by θ trained on a dataset $D = (x_i, y_i)_{i=1}^n$, where $x_i \in R^d$ and y_i is the label. Let X_r represent the feature or label to be forgotten. The classical influence function estimates the effect of removing a data point $z = (x_i, y_i)$ on the model parameters θ as:

$$I(z) = -H_\theta^{-1} \nabla_\theta L(z, \theta), \quad (4)$$

where the Hessian is

$$H_\theta = \frac{1}{n} \sum_{i=1}^n \nabla_\theta^2 L(z_i, \theta). \quad (5)$$

To ensure effective unlearning, an independence constraint is imposed between the model output and the feature to be removed (18):

$$f_\theta(x) \perp X_r. \quad (6)$$

The parameter update is performed iteratively as:

$$\theta' = \theta + \eta \left(-H_\theta^{-1} \sum_{z \in D_r} \nabla_\theta L(z, \theta) + \lambda \nabla_\theta \mathcal{R}_{\text{indep}}(\theta, X_r) \right) \quad (7)$$

where D_r is the set of points containing X_r , η is a step size, λ is a regularization weight, and $\mathcal{R}_{\text{indep}}$ enforces independence from the removed feature.

A more computationally efficient variant of influence-function unlearning is introduced in (20), addressing the high cost of explicitly inverting the Hessian. This method, termed Influence Approximation Unlearning (IAU), uses a gradient-based anti-memorization update. Let D_f denote the data to be forgotten. The update rule is:

$$\theta^{(t+1)} = \theta^{(t)} + \eta \sum_{z \in D_f} \nabla_\theta L(z, \theta^{(t)}) \quad (8)$$

$$- \eta \alpha \sum_{z \in D \setminus D_f} \nabla_\theta L(z, \theta^{(t)}), \quad (9)$$

where η is a step size and α is a correction factor ensuring that performance on retained data remains stable. The updates continue until the influence of D_f falls below a threshold ϵ , providing a scalable alternative to Hessian-based methods.

Influence-function unlearning has also been extended to Bayesian Personalized Ranking (BPR) in recommender systems (2). Let $D_f = (u, i^+)$ represent the positive user–item interactions to forget, with negative items i^- sampled for each user u . The BPR loss is defined as:

$$L_{BPR}(\theta; u, i^+, i^-) = -\log \sigma(f_\theta(u, i^+) - f_\theta(u, i^-)), \quad (10)$$

where $\sigma(\cdot)$ is the sigmoid function. The influence of forgetting a positive interaction (u, i^+) is:

$$I(u, i^+) = -H_\theta^{-1} \sum_{i^- \in N_u} \nabla_\theta L_{BPR}(\theta; u, i^+, i^-), \quad (11)$$

where N_u denotes the negative samples. Model parameters are updated using:

$$\theta' = \theta + \sum_{(u,i^+) \in D_f} I(u, i^+), \quad (12)$$

with adjustments to preserve ranking quality for the remaining interactions.

Collectively, these studies demonstrate that influence-function-based unlearning is highly adaptable across domains. Work in (18) extends unlearning to feature- and label-level removal, the approach in (20) introduces a scalable gradient-based approximation, and the method in (2) adapts influence-based unlearning to structured pairwise ranking tasks. Across all variants, unlearning is performed through carefully designed parameter updates that remove targeted information while preserving overall model utility.

6 Challenges and Research Gaps

The following Table 1 highlights key limitations of FATS and IFU in federated unlearning. FATS struggles with extreme non-IID data, requires storing historical updates, and only supports single-client unlearning (Ref. 26). IFU faces high computational cost, approximation instability, and poor scalability for large models (Refs. 2, 18, 20). These gaps affect both model reliability and privacy guarantees. Addressing them is essential for practical deployment in healthcare federated learning environments.

7 Proposed System/Application

Based on the insights gathered from the literature review and the identified challenges and research gaps in existing federated unlearning techniques, this project proposes a Conceptual Description and Motivation

The proposed system is a Hybrid Federated Unlearning Framework designed to enable hospitals to collaboratively train AI models on sensitive patient data while providing the ability to securely and efficiently remove specific patient information upon request. In healthcare, regulatory frameworks such as GDPR and HIPAA grant patients the right to be forgotten (RTBF), which presents a challenge for conventional federated learning models because model parameters inherently encode contributions from all training data.

Motivated by the need for privacy, compliance, and operational efficiency, our framework integrates FATS (Federated Averaging with Training

Stabilization) for shard-level or hospital-level unlearning, and Influence Function-Based Unlearning (IFU) for precise sample-level deletion. This hybrid approach allows for accurate removal of patient data without full model retraining, reducing computational costs while preserving model utility. It is particularly suited for multi-hospital environments with heterogeneous data distributions, addressing the challenges of non-IID data and multi-client deletion requests.

The primary goal is to provide a scalable, privacy-preserving, and verifiable unlearning solution for healthcare AI applications, ensuring compliance with legal and ethical requirements while maintaining model performance.

7.1 System Overview and Architecture

The proposed hybrid federated unlearning framework is designed to facilitate secure, privacy-preserving, and verifiable removal of patient data in multi-hospital collaborations. The system supports two levels of unlearning granularity: client-level (hospital or shard) and patient-level (individual sample), enabling flexible responses to data deletion requests.

At the client level, entire datasets from a hospital can be removed when an institution withdraws from the federated network or requests bulk deletion. This functionality leverages FATS (Federated Averaging with Training Stabilization) to remove the influence of all local updates from the global model without full retraining.

At the patient level, individual contributions are selectively forgotten using Influence Function-Based Unlearning (IFU). The system computes the influence of each patient’s data on the model parameters and adjusts the global model accordingly, ensuring accurate deletion while preserving overall model performance.

Each hospital acts as a local training node, performing model updates on-site without sharing raw patient data. The federated coordinator aggregates these updates securely using FedAvg or FedProx, maintaining a synchronized global model. When deletion requests are received, the hybrid unlearning engine determines the appropriate strategy, applies corrections, and forwards the updated model to all hospitals.

To guarantee transparency, accountability, and regulatory compliance, the system incorporates a verification and compliance module, which con-

Technique	Gap / Challenge	Explanation	Ref.
FATS	Extreme non-IID sensitivity	TV-stability weakens under severe hospital heterogeneity, reducing stability and accuracy.	(26)
FATS	Requires storing historical updates	Exact unlearning requires retaining all past client updates, increasing storage cost and exposing privacy-sensitive gradients.	(26)
FATS	Only supports single-client unlearning	Cannot handle multi-client deletion or frequent patient RTBF requests in real deployments.	(?)
IFU	Hessian inversion cost	Classical IFU requires expensive Hessian computation and inversion, impractical for deep federated models.	(18)
IFU	Approximation instability	Approximate methods (e.g., IAU) may under-forget or over-forget depending on curvature or noise.	(20)
IFU	Poor scalability to large models	Influence estimates become unreliable for high-dimensional CNN/ViT models used in healthcare FL.	(2)

Table 1: Key gaps and challenges across FATS, and IFU, unlearning methods.

ducts proof-of-unlearning (PoU), divergence analysis, membership inference resistance checks, and audit logging.

Overall, this framework provides a practical and scalable solution for healthcare federated learning, enabling hospitals to maintain high model utility while adhering to privacy regulations and handling heterogeneous data distributions effectively.

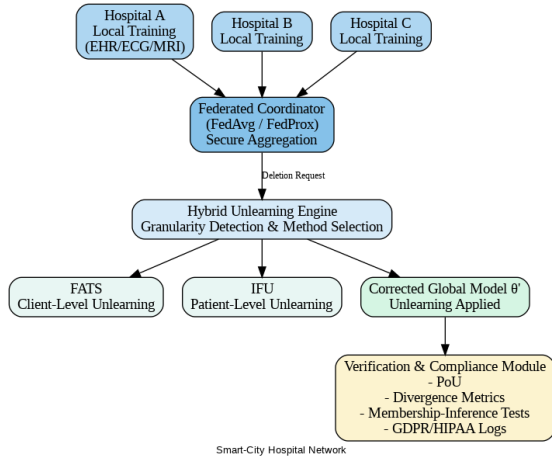


Figure 8: FATS - IFU proposed Hybrid system architecture

7.2 Target application

The proposed hybrid federated unlearning system is designed for smart-city hospital networks,

where multiple healthcare institutions collaboratively train AI models using sensitive patient data such as electronic health records (EHRs), ECG signals, and MRI scans. The target application focuses on enabling hospitals to comply with Right-to-Be-Forgotten (RTBF) requests, data deletion mandates, or patient-specific privacy regulations, without requiring full model retraining.

7.3 Use-Case Scenario: Patient-Specific Unlearning in Smart-City Hospitals

Consider a *smart-city hospital network* consisting of multiple hospitals (Hospital A, B, and C) collaboratively training an AI model for patient diagnosis and risk prediction using sensitive medical data such as EHRs, ECG signals, and MRI scans. A typical scenario arises when a patient requests data deletion; for example, Patient 3 at Hospital B exercises their *Right-to-Be-Forgotten (RTBF)*.

- 1. Deletion Request Submission:** Hospital B receives the deletion request and forwards it to the *Federated Coordinator*. The request specifies that only Patient 3's data should be removed, not the entire hospital dataset.
- 2. Granularity Detection:** The *Hybrid Unlearning Engine* evaluates the request to determine its granularity. Since this is *patient-level*, the system identifies the appropriate unlearning

method as *Influence Function-Based Unlearning (IFU)*.

3. **Patient Influence Computation:** IFU approximates how Patient 3's data has influenced the global model parameters. This involves computing the gradient of the model loss with respect to the patient's data and estimating the impact via Hessian-vector products.
4. **Parameter Update and Unlearning:** Using the influence estimates, the engine removes Patient 3's contribution from the model parameters without retraining the entire model. This preserves the utility of the global model while ensuring that Patient 3's data no longer affects predictions.
5. **Verification and Compliance:** After unlearning, the system verifies the correctness and privacy of the operation, including:
 - Proof-of-Unlearning (PoU) to confirm that Patient 3's influence has been removed.
 - Divergence Metrics to ensure the model's overall predictions remain stable.
 - Membership-Inference Resistance tests to prevent leakage of deleted data.
 - Audit Logs for GDPR/HIPAA compliance.
6. **Updated Model Deployment:** The corrected global model is then available for inference, fully unlearning Patient 3's data while retaining knowledge from other patients.

This scenario demonstrates the patient-level unlearning capability of the proposed system. For client-level requests, e.g., if an entire hospital wants to withdraw all data, the system seamlessly applies *FATS* to remove the corresponding shard, supporting scalable multi-client unlearning. Overall, this use-case highlights the system's real-world applicability, allowing hospitals to meet privacy regulations, handle heterogeneous data, and maintain model performance efficiently.

7.4 Addressing Limitations of Existing Unlearning Techniques

Existing federated unlearning methods, such as *FATS* and *IFU*, exhibit notable limitations when

applied in realistic healthcare scenarios. *FATS* is highly sensitive to extreme non-IID distributions across hospitals, leading to weakened stability and reduced model accuracy (?). It also requires storing all historical client updates to perform exact unlearning, which increases storage costs and may expose privacy-sensitive gradient information (?). Moreover, *FATS* supports only single-client unlearning, making it unsuitable for frequent patient-level deletion requests in real-world deployments.

Classical *IFU*, on the other hand, relies on expensive Hessian computation and inversion, which becomes impractical for deep models common in healthcare federated learning (?). Approximate *IFU* methods can suffer from under-forgetting or over-forgetting depending on model curvature and noise (?), and influence estimates become unreliable for high-dimensional architectures such as CNNs or ViTs (?).

The proposed hybrid system directly addresses these challenges by combining *FATS* and *IFU* in a **granularity-aware framework**. For patient-level deletion requests, the system applies Influence Function-Based Unlearning to selectively remove the patient's impact on the model, avoiding full retraining while preserving utility. For client- or hospital-level deletion, *FATS* is used to remove entire shards efficiently, supporting multi-client unlearning in heterogeneous hospital networks. To further enhance stability, the system uses adaptive optimization techniques and verification modules, including Proof-of-Unlearning, divergence metrics, and membership-inference resistance tests. This approach reduces storage overhead, ensures compliance with privacy regulations, and maintains high model performance across diverse hospital environments.

By integrating patient-level precision and client-level scalability, the proposed system provides a practical, secure, and efficient solution for federated unlearning in real-world healthcare applications.

7.5 Expected Benefits and Limitations

Our proposed hybrid federated unlearning system offers several anticipated benefits. First, it enables precise patient-level unlearning via Influence Function-Based Unlearning, allowing healthcare institutions to comply with patient data deletion requests without full retraining. Second, it supports client-level unlearning through *FATS*, provid-

ing scalability for multi-hospital networks and frequent deletion requests. The adaptive optimization and verification modules enhance stability, prevent under- or over-forgetting, and ensure regulatory compliance (GDPR/HIPAA). Overall, the system is expected to improve model utility, privacy preservation, and operational efficiency in heterogeneous hospital environments.

However, potential limitations remain. Influence function approximation may still introduce minor estimation errors, particularly in highly complex models. Large-scale deployments with very high-dimensional data may incur computational overhead despite the hybrid approach. Additionally, ensuring consistent performance across highly non-IID datasets and real-time deletion requests may pose practical challenges. These limitations indicate areas for future refinement and optimization.

7.6 Future Research Directions

Future work could focus on extending the system toward real-world deployment in large healthcare networks. Suggested directions include:

- Developing more efficient Hessian approximation techniques to reduce computational cost for very deep models.
- Investigating adaptive hybrid strategies that dynamically select between patient- and client-level unlearning based on request frequency and data distribution.
- Integrating federated differential privacy and secure multiparty computation to further enhance data protection.
- Evaluating the framework under diverse hospital scenarios with extreme non-IID distributions and heterogeneous data types (EHR, imaging, genomics).

These open research questions aim to guide the broader community toward practical, secure, and scalable federated unlearning solutions.

7.7 Conclusion

In this work, we surveyed current federated unlearning techniques and highlighted critical gaps in FATS and IFU, including scalability, computational cost, stability, and multi-client limitations. To address these issues, we proposed a hybrid framework combining client-level FATS and patient-level Influence Function-Based Unlearning, aug-

mented with adaptive optimization and verification modules. The system enables precise, scalable, and privacy-compliant unlearning in heterogeneous hospital environments. Our proposed approach provides a foundation for real-world healthcare applications, while outlining clear directions for future research to improve efficiency, robustness, and regulatory compliance.

Authorship Statement

The project presented in this paper was conducted individually by the authors, Amani A., Farha H., as part of the coursework requirements. All conceptualization, system design, implementation, analysis, and documentation were completed solely by the authors.

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